

# LAB REPORT

1)

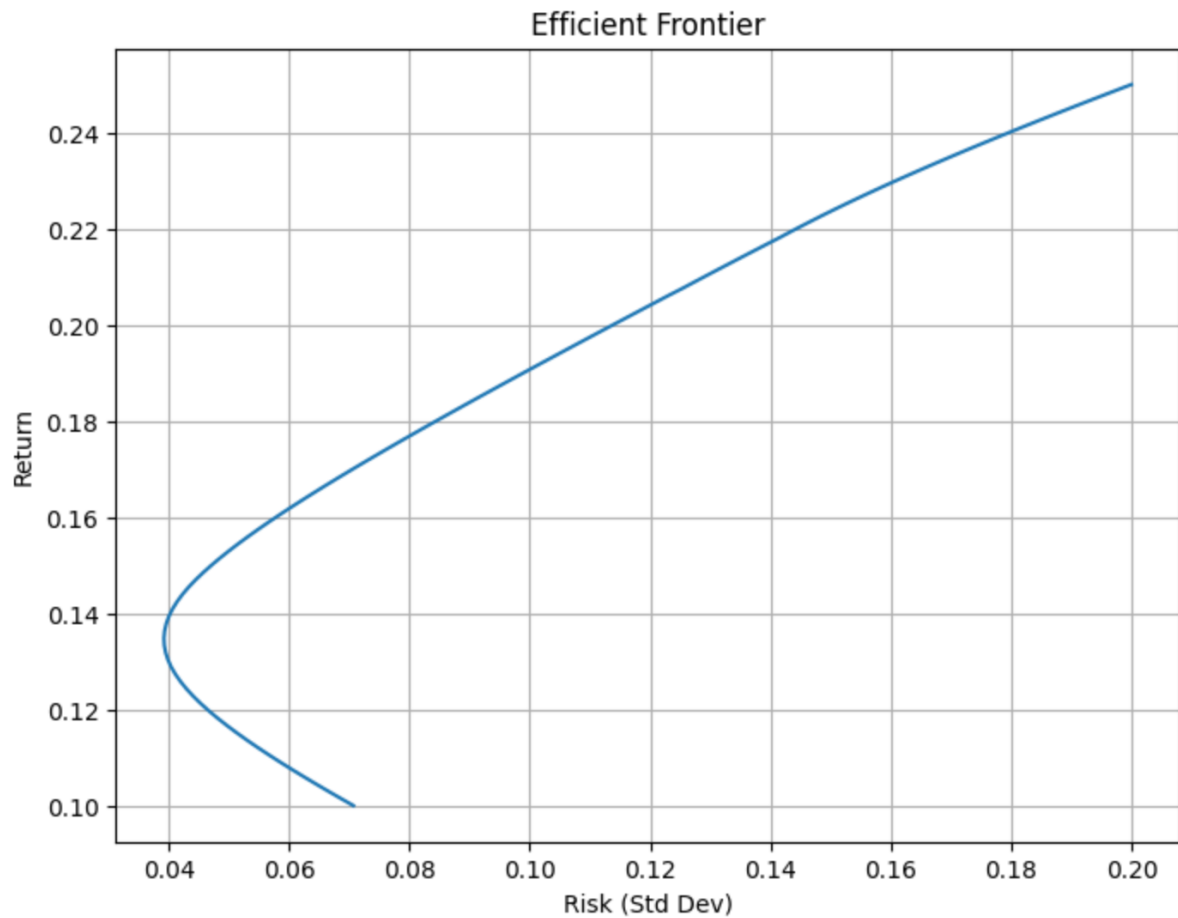


Table of 10 Efficient Portfolios:

|   | w1       | w2       | w3           | Return   | Risk     |
|---|----------|----------|--------------|----------|----------|
| 0 | 1.000000 | 0.000000 | 7.450581e-09 | 0.100000 | 0.070711 |
| 1 | 0.888889 | 0.111111 | 0.000000e+00 | 0.116667 | 0.049690 |
| 2 | 0.777778 | 0.222222 | 0.000000e+00 | 0.133333 | 0.039284 |
| 3 | 0.659686 | 0.329843 | 1.047119e-02 | 0.150000 | 0.047116 |
| 4 | 0.504363 | 0.418848 | 7.678884e-02 | 0.166667 | 0.066189 |
| 5 | 0.349040 | 0.507853 | 1.431065e-01 | 0.183333 | 0.089221 |
| 6 | 0.193717 | 0.596859 | 2.094241e-01 | 0.200000 | 0.113834 |
| 7 | 0.052286 | 0.692810 | 2.549048e-01 | 0.216667 | 0.139224 |
| 8 | 0.000000 | 0.833333 | 1.666667e-01 | 0.233333 | 0.166916 |
| 9 | 0.000000 | 1.000000 | 0.000000e+00 | 0.250000 | 0.200000 |

15% Risk Portfolios:

Max Return: 0.22361309906228566 Weights: [8.82100111e-18 7.36130991e-01 2.63869009e-01]  
Min Return: 0.21979706527667933 Weights: [0.2013529 0.7986471 0.]

### 18% Return Portfolio:

Weights: [0.38010471 0.49005236 0.12984293]

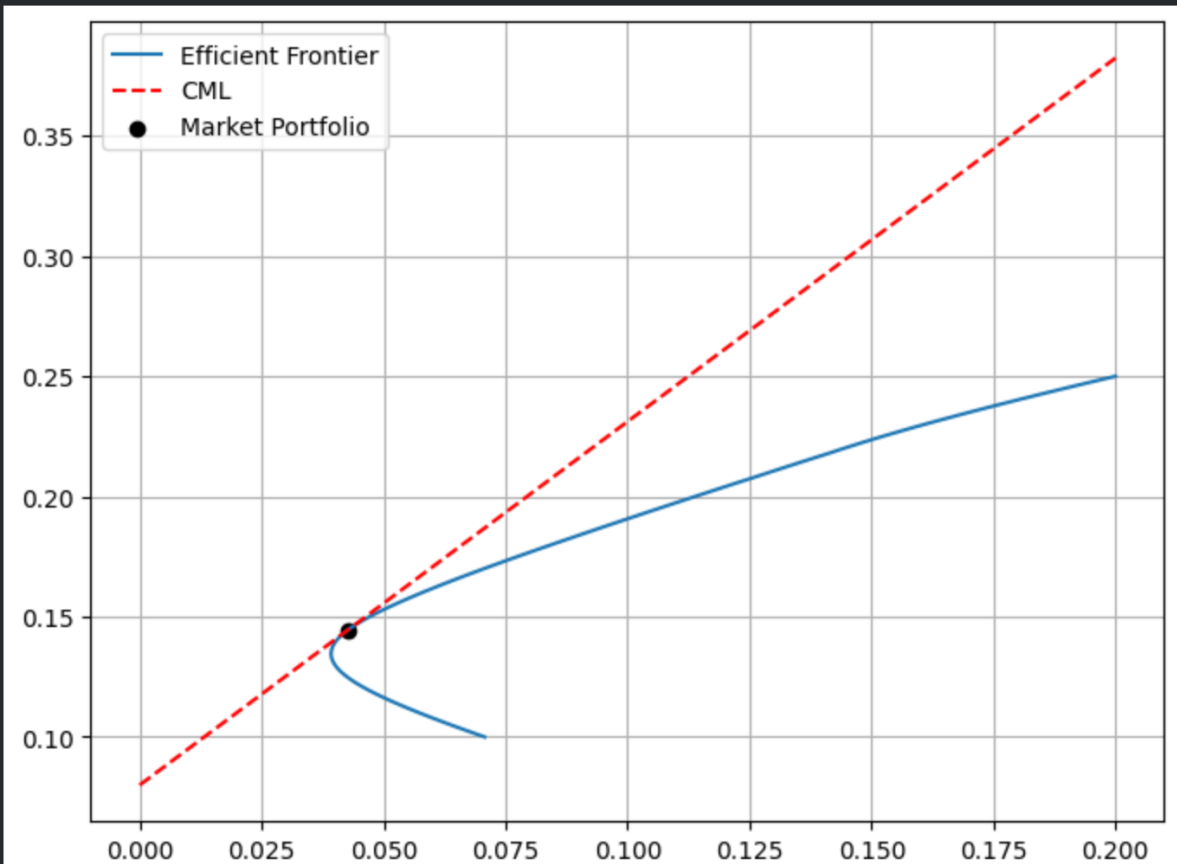
Risk: 0.08444583757351257

### Market Portfolio:

Weights: [7.04218556e-01 2.95781444e-01 1.17943073e-16]

Return: 0.14436721662809462

Risk: 0.04258156586529074

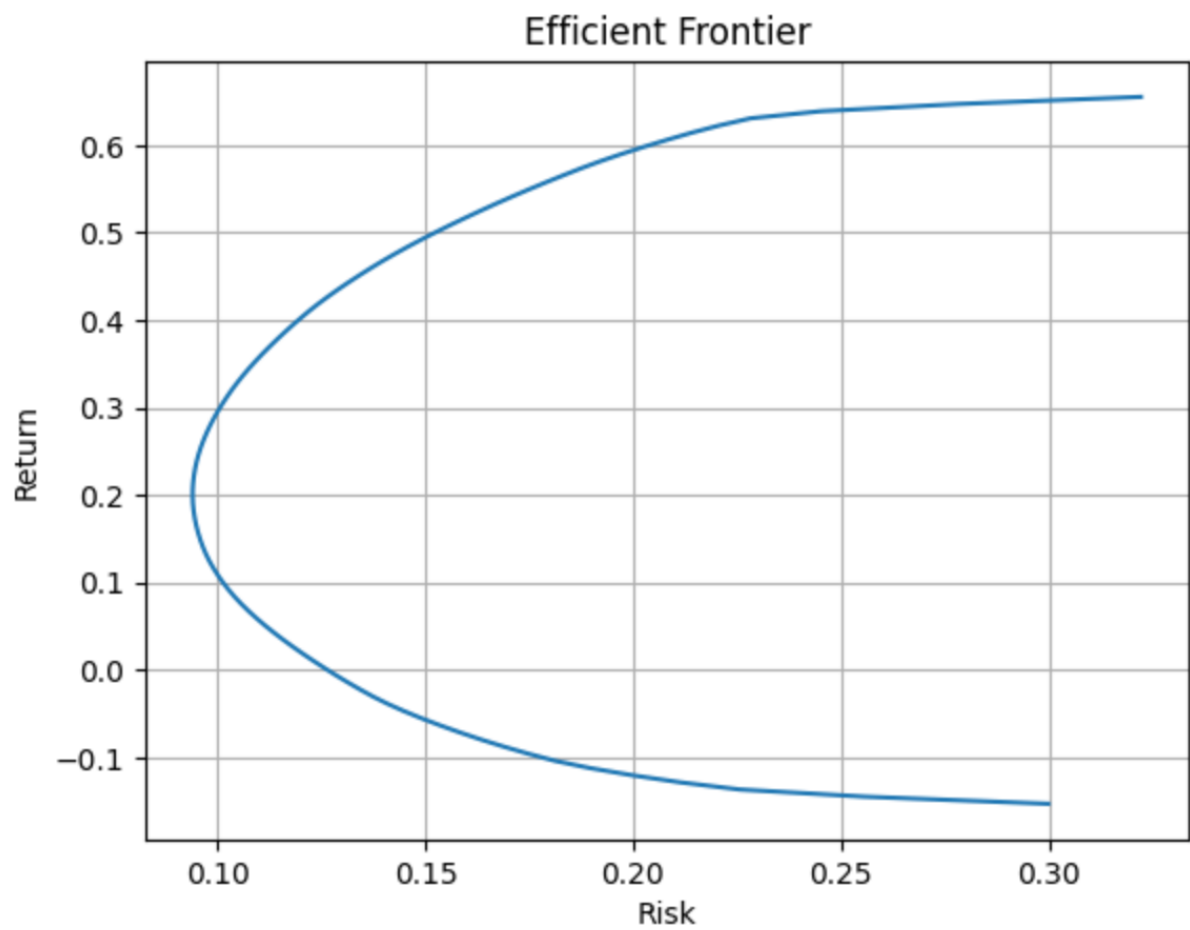


### Portfolios with Risk-Free Asset:

10% Risk Portfolio (risky weights): [1.65381085e+00 6.94623221e-01 2.76981532e-16]

25% Risk Portfolio (risky weights): [4.13452712e+00 1.73655805e+00 6.92453829e-16]

2)



Market Portfolio Weights:

```
Stock_1      7.190523e-02
Stock_2      1.084922e-01
Stock_3      0.000000e+00
Stock_4      2.506844e-01
Stock_5      2.056112e-01
Stock_6      7.309574e-03
Stock_7      5.301165e-16
Stock_8      1.328845e-01
Stock_9      8.690794e-02
Stock_10     1.362049e-01
dtype: float64
```

